Number Sequences (Set B)

Year 4 · **Lesson Worksheet**

Worked Examples

Example 1 — Arithmetic Sequence: What comes next? 4, 9, 14, 19, __

Answer: 24 — each term increases by 5 (add 5 each time).

Example 2 — Geometric Sequence: What comes next? 2, 6, 18, 54, __

Answer: 162 — each term is multiplied by 3.

Example 3 — Two-Step Rule: What comes next? 1, 3, 7, 15, 31, __

Answer: 63 — each term is double the previous term, plus 1 (×2, +1).

Spotting the Pattern (15 questions)

1. What comes next? 3, 6, 9, 12, __
2. What comes next? 50, 45, 40, 35, __
3. What comes next? 2, 4, 8, 16, __ A) 20 B) 24 C) 28 D) 32 E) 36
4. Select ALL that apply: which of these sequences increase by exactly 4 each time? A) 2, 6, 10, 14 B) 5, 9, 13, 18 C) 1, 5, 9, 13 D) 3, 7, 12, 16 E) 10, 14, 18, 22
5. What comes next? 7, 14, 21, 28, __
6. What comes next? 100, 90, 80, 70, __ A) 50 B) 55 C) 60 D) 65 E) 75
7. What comes next? 1, 4, 9, 16, 25, __ (these are square numbers)
8. Fill in the missing number: 5, 10, __, 20, 25 (increasing by 5 each time)
9. What comes next? 1, 2, 4, 8, 16, __ A) 20 B) 24 C) 28 D) 32 E) 36
10. Select ALL that apply: which of these are true about the sequence 3, 9, 27, 81? A) Each term is multiplied by 3. B) Each term increases by 6. C) The next term would be 243. D) The sequence is geometric. E) The sequence is arithmetic.
11. What comes next? 2, 5, 8, 11, __
12. What comes next? 1, 1, 2, 3, 5, 8, __ (the Fibonacci sequence — add the two previous terms) A) 11 B) 12 C) 13 D) 14 E) 15
13. What comes next? 60, 54, 48, 42, __
14. Fill in the missing number: 2, 4, 8, __, 32 (each term doubles)
15. Select ALL that apply: which statements about the sequence 1, 4, 9, 16, 25 are true? A) These are the first five square numbers. B) The next term is 36. C) Each term increases by a constant amount. D) The gap between terms increases each time. E) The sequence is decreasing.

Applying the Pattern (15 questions)

1. What comes next? 4, 7, 12, 19, 28, __ (the gap increases by 2 each time: +3, +5, +7, +9…)

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2. What comes next? 100, 50, 25, 12.5, __

3. What comes next? 2, 3, 5, 8, 13, __ (Fibonacci-style: add the two previous terms)

4. A sequence follows the rule "double, then subtract 1". Starting at 3, what is the 4th term? (3, 5, 9, __)

5. Select ALL that apply: which of these sequences follow a Fibonacci-style rule (each term is the sum of the two before it)? A) 1, 1, 2, 3, 5, 8 B) 2, 4, 6, 8, 10 C) 2, 2, 4, 6, 10, 16 D) 1, 2, 3, 4, 5, 6 E) 3, 5, 8, 13, 21, 34

6. What comes next? 5, 8, 14, 23, 35, __ (the gap increases by 3 each time: +3, +6, +9, +12...)

7. A sequence follows the two-step rule "multiply by 2, then add 3". Starting at 1, what are the first four terms?

8. What comes next in this sequence of triangular numbers? 1, 3, 6, 10, 15, __

9. In the sequence 3, 6, 11, 18, 27, __, what is the rule linking consecutive terms, and what comes next?

10. Select ALL that apply: which of these sequences have a gap that increases by the same amount each time (i.e. the gaps themselves form an arithmetic sequence)? A) 1, 2, 4, 7, 11 (gaps 1,2,3,4) B) 2, 4, 6, 8, 10 (gaps all 2) C) 1, 3, 6, 10, 15 (gaps 2,3,4,5) D) 5, 10, 20, 40 (gaps 5,10,20) E) 2, 5, 9, 14, 20 (gaps 3,4,5,6)

11. What comes next? 200, 100, 50, 25, __

12. A sequence starts 2, 5 and each new term is found by adding the previous two terms. What are the next three terms after 2, 5?

13. What comes next? 1, 2, 6, 24, 120, __ (hint: each term is multiplied by an increasing amount: ×2, ×3, ×4, ×5...)

14. Select ALL that apply: which of these are geometric sequences (each term multiplied by the same constant ratio)? A) 4, 8, 16, 32 B) 3, 6, 9, 12 C) 100, 50, 25, 12.5 D) 2, 6, 18, 54 E) 1, 3, 5, 7

15. A sequence follows the rule "subtract 2, then multiply by 2". Starting at 10, find the next two terms after 10.

Sequences in Context (15 questions)

1. What comes next in this sequence, and what two-step rule is being used? 2, 5, 11, 23, ?

2. A sequence is defined by: term = (previous term × 2) + 1. If the first term is 2, find the first four terms of the sequence.

3. What comes next in this Fibonacci-style sequence (each term is the sum of the two before it)? 1, 1, 2, 3, 5, ? A) 6 B) 7 C) 8 D) 9 E) 10

4. What comes next in this sequence of triangular numbers? 1, 3, 6, 10, ?

5. Select ALL that apply: for the sequence 3, 6, 12, 24, which of these are true? A) Each term is multiplied by 2. B) Each term increases by 3. C) The next term would be 48. D) The sequence is geometric. E) The sequence is arithmetic.

6. A sequence follows the rule "double the number, then subtract 2". Starting at 4, what is the 3rd term? (4, 6, __)

7. What comes next? 2, 4, 7, 11, 16, ? (the gap increases by 1 each time: +2, +3, +4, +5...) A) 20 B) 21 C) 22 D) 23 E) 24

8. A sequence starts 1, 2 and each new term is found by adding the previous two terms. What are the next three terms after 1, 2?

9. What comes next in this sequence, which alternates two simple operations: 5, 9, 7, 11, 9, ? (hint: alternately add 4 and subtract 2)

10. A student claims the sequence 2, 4, 8, 16 is arithmetic. Explain why this is incorrect, and state what type of sequence it actually is.

11. What comes next? 5, 10, 20, 40, ?

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12. A sequence starts 2, 3 and each new term is found by adding the previous two terms. What are the next three terms after 2, 3?

13. A sequence follows the rule "subtract 3, then multiply by 2". Starting at 10, what is the next number in the pattern? A) 12 B) 14 C) 16 D) 18 E) 20

14. Select ALL that apply: which of these are geometric sequences (each term multiplied by the same constant ratio)? A) 2, 4, 8, 16 B) 3, 6, 9, 12 C) 60, 30, 15, 7.5 D) 5, 10, 20, 40 E) 1, 3, 5, 7

15. A sequence follows the rule "add 4, then multiply by 2". Starting at 1, find the next two terms after 1.

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